Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 37: Approved Solutions

Graphs, Degree, and the First Theorem

Covers Chartrand & Zhang, Section 12.1

Status of this set

Ungraded practice. Solutions released with it. **Draw each graph** before answering — a sketch catches most errors.

Part I: How to Practice This Material

The one tool

$$\sum_{v \in V(G)} \deg v = 2m$$

Everything in this set follows from it. Two consequences:

- The degree sum is always **even**.
- The number of **odd** vertices is always **even**.

Three mistakes to expect

1. **Forgetting the factor of 2.** The sum is $2m$, not m.
2. **Accepting an odd degree sum.** No such graph exists.
3. **Assuming a legal degree sum means the graph exists.** It is necessary, not sufficient.

Part II: Practice Problems

Section A — Reading a graph

1. Let $V = \{v_1, \ldots, v_6\}$ and $E = \{v_1v_2, v_1v_3, v_2v_3, v_3v_4, v_4v_5\}$. Draw the graph, then give its order, size, degree sequence, and the number of isolated vertices.

Approved solution

Order 6. **Size** 5.
Degrees: $\deg v_1 = 2$, $\deg v_2 = 2$, $\deg v_3 = 3$, $\deg v_4 = 2$, $\deg v_5 = 1$, $\deg v_6 = 0$. Sequence: $0, 1, 2, 2, 2, 3$.

Isolated: one, namely v_6.
Check: the degrees sum to $10 = 2 \cdot 5$. ✓

v_6 appears in V but in no edge — a vertex set may contain vertices no edge touches, and missing that is the standard slip when reading from a list.

Section B — Using the degree sum

2. A graph has size 21. It has two vertices of degree 5, three of degree 6, and all remaining vertices have degree 2. What is its order?

Approved solution

Let x be the number of degree-2 vertices. By Theorem 12.10,

$$2(5) + 3(6) + 2x = 2(21) = 42,$$

so $28 + 2x = 42$ and $x = 7$. The order is $2 + 3 + 7 = 12$.
Check: $10 + 18 + 14 = 42 = 2 \cdot 21$. ✓

3. For each, decide whether a graph with the stated degrees exists. Justify.

(a) $2, 2, 3, 3, 4$

(b) $1, 1, 2, 3, 3, 4$

(c) $1, 2, 2, 3, 3, 4$

(d) A 4-regular graph of order 7

Approved solution

(a) Possible. Sum 14, even, so $m = 7$. Two odd vertices — an even count.
(b) Possible. Sum 14, so $m = 7$. Odd degrees are $1, 1, 3, 3$ — four of them, an even count.
(c) Impossible. Sum $1 + 2 + 2 + 3 + 3 + 4 = 15$, which is odd and cannot equal $2m$. Equivalently, the odd degrees are $1, 3, 3$ — three of them, contradicting Corollary 12.12.
(d) Impossible. The degree sum would be $7 \cdot 4 = 28$, giving $m = 14$ — that part is fine. But a 4-regular graph of order 7 *does* exist (each vertex adjacent to 4 of the other 6). **So (d) is possible**, with $m = 14$.

Note the contrast between (c) and (d). In (c) the parity test rules the graph out immediately. In (d) the parity test passes and the graph does exist — but passing the test is never by itself a proof of existence. Compare a 3-regular graph of order 5: the sum 15 is odd, so that one is impossible.

Section C — Regular graphs

4. For which orders n does a 3-regular graph exist? Justify.

Approved solution

The degree sum is $3n$, which must equal $2m$ and therefore be even. Since 3 is odd, $3n$ is even exactly when n is **even**.

Also $n \geq 4$ is needed, since each vertex requires 3 distinct neighbours.
So a 3-regular graph exists exactly for even $n \geq 4$. For $n = 4$ it is K_4; for $n = 6$, the complete bipartite graph $K_{3,3}$ works.

The general rule: an r-regular graph of order n requires nr even, so n and r cannot both be odd.

Section D — Writing

5. In four to six complete sentences, prove the First Theorem of Graph Theory and explain why it is a counting argument rather than an algebraic one.

Approved solution

Model answer. Consider the collection of all pairs consisting of a vertex together with an edge incident to it. Counting these pairs by vertex gives the sum of the degrees, since each vertex v contributes exactly $\deg v$ pairs. Counting the same pairs by edge gives $2m$, since each edge has exactly two endpoints and so contributes exactly two pairs. The two counts are of the same collection, so they are equal, and $\sum \deg v = 2m$. This is a counting argument because nothing is manipulated algebraically — we simply describe one set of objects in two ways and conclude the descriptions agree.

Full credit requires naming what is being counted twice. An answer that says only "each edge has two ends" is close but should make the double count explicit.

Section E — Challenge (optional)

6. Show that in any graph of order $n \geq 2$, some two vertices have the same degree.

Approved solution

Pigeons: the n vertices. **Holes:** the possible degrees $0, 1, \ldots, n-1$ — that is n holes, so pigeonhole does not apply yet.
The observation: degrees 0 and $n-1$ cannot both occur. A vertex of degree $n-1$ is adjacent to every other vertex, so no vertex can have degree 0; conversely a vertex of degree 0 prevents any vertex from reaching degree $n-1$.
So at most $n-1$ degrees are available for n vertices, and two must coincide. ■

This is the handshake problem from Practice Set 33, now stated in the language it was always about. "Number of acquaintances" is degree, and "group of people" is a graph.

Instructor notes

- **Problem 3(c) versus 3(d)** is the point of the section: a failed parity test settles the matter, a passed one does not.
- **Problem 1's isolated vertex** catches students who read only the edge list.
- **Problem 6** is Practice Set 33 Problem 6 restated in graph language. Showing the two side by side makes the modelling step visible.
- **Insist on sketches.** Students who draw catch far more errors than those who manipulate lists.

- Reserved for Homework 14: Section 12.1 even-numbered exercises. Odd ones have printed answers and are self-check only.